Maths IA-1 — The Only Questions That Matter

Chapter-wise · topic-wise · high-yield shortlist for 26 Oct 2026 · answers inline

Built from Contest-1's real blueprint (18 Sep, 35M/90 min) + central-team twist style. **If you do only this sheet, you have trained every topic the paper can build from.**

STRATEGY CARD — 35 marks / 90 minutes

- **Section A** — 5 MCQ × 1M (5 min): reasoning-based, not calculation. Spot the flawed logic, then eliminate.
- **Section B** — 5 one-mark values (7 min): sign, units, "rejected root" traps decide these.
- **Section C** — 1 match × 5M (7 min): do each pair independently; never chain-guess.
- **Section D** — 5 × 4M = **20M (60 min)**: the paper is won here. 4 marks = show every step + interpret in words. No steps = 1 mark even if answer is right.
- Buffer 10 min: re-check Section B signs and log rejections.

Mark math: 15 easy marks (A+B+C) + 20 working. Bank the 15 first, then grind D.

CHAPTER 1 — Algebraic Expressions (expect ~6M · heaviest in Contest-1)

Topics: expressions from words · area models · identities · x ± 1/x family · exponent rules · factorisation

1. **[B]** A square of side x has four 1×1 corners removed. Remaining area = ? → **$x^2 - 4$**

2. **[B]** If $x + 1/x = 5$ then $x^2 + 1/x^2$ = ? → **23** *(use $(x+1/x)^2 - 2$)*

3. **[A]** Price p, sales = 20,000 − 500p. Revenue = ? → **p(20,000 − 500p)** *(quantity × price; the "+p / −p" options are the bait)*

4. **[B]** Simplify $(x^2y^3)^4 \div (x^3y^5)$ → **x^5y^7**

5. **[D]** Saira has one square x×x, eight strips x×1, and fifteen 1×1 squares. (i) total area = ? (ii) arrange into ONE rectangle — length & breadth? → **(i) $x^2 + 8x + 15$ (ii) (x+5)(x+3): sides x+5 and x+3**

6. **[D]** Growing-square pattern: steps have 1, 4, 9 squares. (i) equation in n? (ii) steps 4 and 15? → **(i) $y = n^2$ (ii) 16 and 225** *(always state WHY: n×n grid at step n)*

7. **[D]** Prove $(2n+1)^2 - (2n-1)^2$ is divisible by 8 for every integer n. → **difference of squares: [(2n+1)+(2n−1)] [(2n+1)−(2n−1)] = (4n)(2) = 8n = 8·n ∎**

8. **[C]** Match: a) $(a+b)^2 - (a-b)^2$ → **4ab** · b) a^2+b^2 if a+b=7, ab=10 → **29** · c) a^3+b^3 if a+b=7, ab=10 → **133** · d) (x+2)(x−5) → **$x^2-3x-10$**

Method cards: $x^2+1/x^2 = (x+1/x)^2 - 2$ · $a^2+b^2 = (a+b)^2 - 2ab$ · $a^3+b^3 = (a+b)^3 - 3ab(a+b)$

CHAPTER 2 — Quadratic Equations (expect ~7M)

Topics: discriminant & root nature · equal-roots k-problems · sum/product of roots · completing the square · quadratics in word problems

9. **[A]** $y = x^2 - 4x + k$ touches the x-axis at exactly one point — which must hold? → **discriminant is zero** *(touching = one repeated root, NOT k=0)*

10. **[B]** $x^2 + (k-3)x + k = 0$ has equal roots. Sum of all possible k = ? → $D = (k-3)^2 - 4k = k^2-10k+9 = 0$ → k = 1 or 9 → **10**

11. **[B]** Write $x^2 + 6x + 2$ in completed-square form → **$(x+3)^2 - 7$**

12. **[B]** $3x^2 - 2x + 5$: discriminant = ? and root nature? → **$4 - 60 = -56 < 0$ → no real roots**

13. **[D]** Rocket height $h(t) = -5t^2 + 20t + 25$: (i) zeros (ii) max height WITHOUT calculus (iii) $h(6) = -35$ — "rocket is 35 m underground"? → **(i) $-5(t^2-4t-5) = -5(t-5)(t+1)$ → t = 5 s (t = −1 rejected, time can't be negative) (ii) vertex $t = -b/2a = 2$ → $h(2) = -20+40+25 = 45$ m (iii) NO — model is valid only during flight; t = 6 s is past landing, negative height is extrapolation**

14. **[D]** If α, β are roots of $2x^2 - 5x - 3 = 0$, find $\alpha^3 + \beta^3$. → sum = 5/2, product = −3/2; $\alpha^3+\beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta) = 125/8 + 45/4 =$ **215/8**

15. **[D]** Find k so that $ax^2 - 2x + 3 = 0$ has BOTH roots positive. → sum $= 2/a > 0$ → $a > 0$; $D = 4 - 12a \geq 0$ → $a \leq 1/3$ → **$0 < a \leq 1/3$** *(state all three: sum>0, product>0, D≥0)*

Method cards: equal roots ⟺ $D = 0$ · $\alpha^2+\beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$ · vertex $t = -b/2a$ · never accept the negative root in time/distance contexts — say why.

CHAPTER 3 — Inequalities (expect ~5M)

Topics: linear flip-traps · quadratic sign · absolute value · rational · chained application · "for all x" conditions

16. **[A]** If $a < b$, which is ALWAYS true? → **$a + 7 < b + 7$** *($-2a < -2b$ flips → false; $a^2 < b^2$ fails for negatives; $1/a > 1/b$ fails without sign info)*

17. **[B]** If $1/x > 0$ then x is → **positive ($x > 0$)**

18. **[B]** Solve: $-3x + 4 \leq 10$ → **$x \geq -2$** *(flip when dividing by −3 — the guaranteed trap)*

19. **[B]** Solve $x^2 - 5x + 6 < 0$ → **$2 < x < 3$** *(negative BETWEEN the roots)*

20. **[D]** IQ = MA/CA × 100. Children aged 15 with $90 \leq IQ \leq 150$: mental-age range? → $MA = IQ \times 15/100$ → **$13.5 \leq MA \leq 22.5$ years**

21. **[D]** Solve $|x - 3| \leq 5$ → **$-2 \leq x \leq 8$** *(write as $-5 \leq x-3 \leq 5$, then add 3 — show the compound-inequality step)*

22. **[D]** Find all real k with $x^2 + 2x + k > 0$ for ALL real x. → $D = 4 - 4k < 0$ → **$k > 1$** *(mention: k = 1 fails because x = −1 gives 0)*

23. **[C]** Match: a) multiply by −1 → **flip sign** · b) $x^2 - 9 < 0$ → **$-3 < x < 3$** · c) $|x| \leq a$ → **$-a \leq x \leq a$** · d) $x^2 + 1 > 0$ for all x → **always true ($D < 0$)** · e) $(x-1)/(x+2) \geq 0$ → **$x \geq 1$ or $x < -2$** *($x = -2$ excluded)*

CHAPTER 4 — Logarithms & Exponents (expect ~5M)

Topics: definition · laws · chain/cycle products · equations + rejection · digit counting

24. **[B]** Evaluate: $\log_2 32 =$ **5** · $\log_9 27 =$ **3/2** · $\log 1000 =$ **3**

25. **[B]** $\log_2 6 - \log_2 3 =$ **1** · $\log 45$ in terms of $\log 3$, $\log 5 =$ **$2\log 3 + \log 5$**

26. **[B]** $\log_2 3 \times \log_3 8 =$ → chain collapses to $\log_2 8 =$ **3** *(favourite: the middle bases cancel)*

27. **[D]** Solve: log x + log(x – 3) = 1 → log[x(x–3)] = 1 → x^2–3x–10 = 0 → x = 5 or –2; **x = 5** *(must write: x = –2 rejected, log of negative undefined)*

28. **[D]** How many digits does 5^{20} have? ($\log_{10}5 = 0.6990$) → $\log(5^{20}) = 13.98$ → floor = 13 → 13 + 1 = **14 digits**

29. **[D]** A colony doubles every 3 hours; starts at 100. Hours to reach 6,400? → $100 \cdot 2^n = 6400$ → $2^n = 64$ → n = 6 doublings × 3 h = **18 hours**

30. **[C]** Match: a) $2^x = 7$ → **$x = \log_2 7$** · b) $\log_8 4$ → **2/3** · c) log 0.01 → **–2** · d) if $a^b = c$ then log_a c → **b** · e) $\log_2 8 + \log_2 4$ → **3 + 2 = 5**

Method cards: log(A/B) = log A – log B · log_p(p) = 1 · change of base a^b = N ⟺ b = log_a N · digits = ⌊log N⌋ + 1 · **always list domain restrictions before accepting roots.**

CHAPTER 5 — Divisibility, GCD & LCM (expect ~6M · rank decider)

Topics: Euclidean algorithm · gcd×lcm · simultaneous conditions · inclusion–exclusion · divisor counting · word problems

31. **[B]** gcd(48, 18) = **6** · lcm(12, 18) = **36** · gcd × lcm = product check: gcd 8, lcm 96, a = 24 → b = 96×8/24 = **32**

32. **[B]** Number of positive divisors of 60 = ? → $60 = 2^2 \cdot 3 \cdot 5$ → (2+1)(1+1)(1+1) = **12**

33. **[D]** Euclidean algorithm: gcd(1071, 462) — write full steps → 1071 = 2×462 + 147; 462 = 3×147 + 21; 147 = 7×21 + 0 → **21** *(steps ARE the marks)*

34. **[D]** Buses leave a stop every 15 min and every 20 min, both at 8:00. Next simultaneous departure? → lcm(15,20) = 60 → **9:00 AM**

35. **[D]** How many integers from 1 to 100 are divisible by 6 OR by 8? → ⌊100/6⌋ = 16, ⌊100/8⌋ = 12, ⌊100/24⌋ = 4 → 16 + 12 – 4 = **24** *(state inclusion–exclusion explicitly)*

36. **[D]** Greatest number that divides 64, 127 and 187 leaving the SAME remainder each time? → N divides all pairwise differences: 63, 60, 123 → gcd = **3** *(check: remainders 1, 1, 1 ✓)*

37. **[D]** $n = 5^2 \cdot 7^3 \cdot 11$. How many positive divisors, and why? → (2+1)(3+1)(1+1) = **24** *(explain: each prime power picks 0…max)*

38. **[C]** Match: a) gcd(48, 18) → **6** · b) lcm(8, 12) → **24** · c) largest divisor of 240, 320, 400 → **80** · d) $2^{21} - 1$ divisible by 7? → **yes ($2^3 \equiv 1$)** · e) divisors of p^2q (p, q prime) → **6**

Method cards: gcd(a,b) × lcm(a,b) = a·b · Euclid: divide, remainder, repeat until 0 · equal-remainder ⇒ divide the DIFFERENCES · τ(n) = product of (exponent + 1) · "div by a or b" ⇒ inclusion–exclusion.

CHAPTER 6 — Congruence & Modular Arithmetic (expect ~6M · rank decider)

Topics: remainder & last digits · cyclicity · big powers mod n · inverses · linear congruences · proofs

39. **[B]** 100 mod 7 = **2** · last digit of 3^{250} = ? → cycle of 3 is (3,9,7,1); 250 mod 4 = 2 → **9**

40. **[B]** Remainder of $4^{50} \div 7$ = ? → $4^3 = 64 \equiv 1 \pmod 7$; 50 = 3×16 + 2 → $4^2 = 16 \equiv$ **2**

41. **[B]** Modular inverse of 3 mod 11 = **4** *(3×4 = 12 ≡ 1)*

42. **[B]** Solve $3x \equiv 6 \pmod 9$: divide by 3 → $x \equiv 2 \pmod 3$ → **x = 2, 5, 8 (mod 9)** *(list all solutions in 0…8 — 3 of them)*

43. **[D]** Last two digits of 3^{2024} = ? → order of 3 mod 100 is 20; 2024 mod 20 = 4 → 3^4 = **81**

44. **[D]** Find 2^{100} mod 7 → $2^3 \equiv 1$; 100 = 33×3 + 1 → 2^1 = **2**

45. **[D]** Prove $n^3 - n$ is always divisible by 3. → $n^3 - n = (n-1)n(n+1)$ = three consecutive integers; one of them is a multiple of 3 → product divisible by 3 ∎ *(write the factorisation — that's the proof)*

46. **[D]** A festival repeats every 23 days; today it falls on a Sunday. Day of week after 200 days? → 200 mod 23 = 200 − 184 = 16 → 16 days after Sunday = 2 full weeks + 2 → **Tuesday** *(184 = 8×23; check: +14 lands on Sunday again, +2 more = Tuesday)*

47. **[C]** Match: a) 7^{222} mod 5 → **4** · b) $x + 2 \equiv 0 \pmod 7$ → **$x \equiv 5 \pmod 7$** · c) last digit of 2^{100} → **6** (cycle 2,4,8,6; 100 mod 4 = 0 → 4th) · d) 100 mod 9 → **1** · e) $n^3 - n$ mod 3 → **0 always**

SECTION-D BATTERY — the 20 marks that decide everything

5 × 4M. Practice these under a 12-min clock, steps visible:

1. **(Ch1+2)** A rectangular garden is 5 m longer than it is wide; area 36 m². Find dimensions, then find width if area were 40 m². → $w(w+5) = 36 \to w^2+5w-36 = 0 \to (w+9)(w-4) = 0 \to$ **w = 4 m, l = 9 m**; for 40: $w^2+5w-40 = 0 \to w = (-5+\sqrt{185})/2 \approx$ **4.3 m** ($\sqrt{185} \approx 13.60$)

2. **(Ch3)** Ticket price ₹p, demand 2000 − 50p; company wants revenue ≥ ₹30,000. Solve the inequality. → $p(2000-50p) \geq 30000 \to 2000p - 50p^2 \geq 30000 \to p^2 - 40p + 600 \leq 0 \to D = 1600 - 2400 < 0$ and leading coeff > 0 → **never possible** *(the "trick answer" is the point — say it in words)*

3. **(Ch4)** $pH = -\log_{10}[H^+]$. Solution A has $[H^+] = 10^{-3}$, B has $10^{-4.5}$. Which is more acidic and by how many pH units? → pH A = 3, B = 4.5 → **A more acidic by 1.5 pH units**

4. **(Ch5)** Three lights blink every 12 s, 18 s, 30 s; all together now. Next simultaneous blink? Count blinks in 10 min (incl. t = 0). → lcm = 180 s; 10 min = 600 s → floor(600/180) = 3 after t=0 → **180 s (3 min)**; total = 3 + 1 = **4 times**

5. **(Ch6)** Find the remainder when 7^{99} is divided by 100 (full method). → mod 4: $7 \equiv -1 \to 7^{99} \equiv -1 \equiv 3$; mod 25: $7^2 = 49 \equiv -1 \to 7^4 \equiv 1 \to 7^{99} = 7^{4 \cdot 24} \cdot 7^3 \equiv 343 \equiv 18$; $x \equiv 3 \pmod 4$, $x \equiv 18 \pmod{25}$ → $x = 18 + 25k$, $18 + k \equiv 3 \pmod 4 \to k = 1 \to$ **43**

LAST 60 MINUTES — everything compressed

- **Ch1:** $(x+1/x)^2 - 2$ · $(a+b)^3 - 3ab(a+b)$ · $4ab$ identity · area = square − pieces
- **Ch2:** $D = b^2-4ac$; touch ⟺ D=0; both roots positive ⟺ sum>0, product>0, D≥0; vertex = $-b/2a$
- **Ch3:** flip ONLY when × or ÷ negative · $|x| \leq a \iff -a \leq x \leq a$ · quadratic negative between roots · reject x = −2 in rational
- **Ch4:** laws, chain collapse, digits = $\lfloor \log \rfloor + 1$, **declare domain before accepting roots**
- **Ch5:** gcd×lcm = product · Euclid steps shown · equal remainder ⟹ gcd of differences · $\tau = \Pi(e_i+1)$ · inclusion–exclusion
- **Ch6:** reduce base mod n, then reduce exponent mod cycle order · inverse = multiply to ≡1 · consecutive-product proofs · last two digits = mod 100
- **Section D rule:** every line of algebra earns marks; interpretation questions need a full sentence answer.
- **Never do:** leave blank (method marks exist), skip "explain why" words, accept a negative time/length root silently.

Sheet ends. If a question here feels easy in the exam — that's the design working. Do the 45, sleep 6 hours, go score 30+.